Area Under Curve: Formula Sheet

Basic area

Curve and x-axis: $A = \int_a^b |y|\,dx$

Curve and y-axis: $A = \int_c^d |x|\,dy$

Two curves: $A = \int_a^b |f(x) - g(x)|\,dx$

Horizontal strips: $A = \int_c^d (x_R - x_L)\,dy$

Standard areas

Circle: πr^2

Ellipse: πab

$y^2 = 4ax$, $x^2 = 4by$:

$$\frac{16ab}{3}$$

$y^2 = 4ax$, $y = mx$:

$$\frac{8a^2}{3m^3}$$

Parabola and line:

$$\frac{|a|(\beta - \alpha)^3}{6}$$

$y^2 = 4ax$ and double ordinate $x = h$:

$$\frac{2}{3}(2h)(2\sqrt{ah}) = \frac{8\sqrt{a}\,h^{3/2}}{3}$$

$|x| + |y| \le a$: $2a^2$

Loop $ay^2 = x^2(a - x)$:

$$\frac{8a^2}{15}$$

One arch of $\sin x$: 2

Shifting and images

Shift: $X = x - h$, $Y = y - k$, area unchanged

Image in $y = x$: $(x, y) \to (y, x)$

Image in $y = -x$: $(x, y) \to (-y, -x)$

Image in $x = a$: $(x, y) \to (2a - x, y)$

Image in $y = b$: $(x, y) \to (x, 2b - y)$

Inverse:

$$\int_a^b f + \int_{f(a)}^{f(b)} f^{-1} = bf(b) - af(a)$$

f and f^{-1}: $A = 2\int |f(x) - x|\,dx$

Variable area

f monotonic, $y = f(c)$: least at $c = \dfrac{a+b}{2}$

$$A(x) = \int_0^x f \Rightarrow f(x) = A'(x)$$

Parabola-line area:

$$\frac{|a|}{6}\left(\sqrt{D}/|a|\right)^3$$

Average and RMS

Average:

$$\frac{1}{b-a}\int_a^b f\,dx$$

RMS:

$$\sqrt{\frac{1}{b-a}\int_a^b f^2\,dx}$$

On $(0, \infty)$:

$$\lim_{b\to\infty}\frac{1}{b}\int_0^b f\,dx$$

Parametric

$$A = \left|\int_{t_1}^{t_2} y\,\frac{dx}{dt}\,dt\right|$$

$x = a\cos t$, $y = b\sin t$: πab